

Activity: Match the Crypto Tool to the Task (optional)

1. Secure Email Communication

A company wants to send confidential financial reports between departments. They need to make sure only the intended recipient can read the email, and that the sender can be verified.

- Best Technique: Asymmetric
- Why? Asymmetric because it is best for secure exchanging data over open networks and email.

2. Verifying Software Downloads

An IT department wants to ensure that downloaded software hasn't been tampered with or corrupted. They don't need to encrypt it—just verify it's intact and authentic.

- Best Technique: Hashing
- Why? Hashing creates fixed-length fingerprints that cannot be reversed. Verifies software downloads and file integrity

3. Encrypting Stored Customer Data

A retailer stores large amounts of customer data (names, addresses, payment details) in a local database. They want to protect this data in case of theft or a system breach.

- Best Technique: Symmetric
- Why? Symmetric encrypts and decrypts with one shared key. Very fast, so it is used for files, hard drives, and bulk data.